Ultimate Financial Risk Analysis Report

Introduction

This project analyzes financial transaction data to understand behavior, detect risks, and support decision-making. The workflow includes cleaning, analysis, segmentation, risk detection, and hypothesis testing.

Task 1: Data Cleaning & Preprocessing

Cleaning ensures the dataset is reliable. Dates are converted, numeric fields corrected, duplicates removed, and missing values handled.

```
df['TransactionDate'] = pd.to_datetime(df['TransactionDate'])
df['TransactionAmount'] = pd.to_numeric(df['TransactionAmount'], errors='coerce')
df.ffill(inplace=True)
df.drop_duplicates(inplace=True)
```

Outcome: Clean, structured dataset ready for analysis.

Task 2: Descriptive Analysis

Monthly Trend

Shows how transaction volume changes over time. Peaks indicate high activity periods.

Credit vs Debit

Compares inflow vs outflow. Helps understand financial balance.

Top/Bottom Accounts

Top accounts are financially strong; bottom accounts indicate risk or overdraft.

Graphs: Line plots (trends), bar charts (ranking).

Task 3: Customer Segmentation

Activity Levels

Customers grouped into high, medium, low activity.

Frequency vs Balance

Scatter plot shows relationship. High frequency + low balance = risky.

```
profile = pd.DataFrame({'freq':freq, 'balance':balance})
```

Task 4: Risk Identification

Large Transactions

Top 5% transactions indicate abnormal activity.

Anomaly Detection

Z-score detects extreme values.

```
df['zscore'] = abs(stats.zscore(df['TransactionAmount']))
anomalies = df[df['zscore'] > 3]
```

Suspicious Accounts

Accounts with repeated anomalies are flagged.

Task 5: Insights

Key findings include transaction trends, customer segmentation, and detection of risky accounts.

Task 6: Hypothesis Testing

T-test checks statistical difference between groups.

```
stats.ttest_ind(high_volume['AccountBalance'], low_volume['AccountBalance'])
```

If p-value < 0.05, difference is significant.

Conclusion

The project demonstrates how data analytics can identify patterns, segment customers, and detect financial risks effectively.